Quantitative Trading Using AI Agents

Transforming quantitative research with intelligent agents

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Lecture 1 Recap: Large Language Models for Trading

The Paradigm Shift

From structured numerical data to processing natural language at scale

From Predictive to Generative

Traditional ML vs. Generative AI and next-token prediction

How LLMs Work

Text encoding → pattern recognition → text decoding

RNNs & LSTMs

Sequential processing with memory gates to handle long-term dependencies

Attention Mechanism

Decoder can access all input data, enabling context understanding across long documents. LSTMs + Attention outperformed only LSTMs

Transformers

Parallel processing, simpler architecture, outperformed LSTM+Attention

LLM Architecture

Pre-training on internet data + Fine-tuning with RLHF

Prompt Engineering

Zero-shot vs. Few-shot, Chain-of-Thought, System Prompts

Real Use Cases

News sentiment analysis, earnings call parsing, research summarization, signal generation

Production Challenges

Hallucinations, latency, model drift, compliance risks, cost at scale

Lecture Agenda: Key Topics

Introduction to Agentic AI

Understanding the core concepts and capabilities.

Evolution of Quantitative Trading

Understanding the progression from manual trading to AI agents

Why Single LLM Calls Fail

Limitations of single-step LLM approaches in trading

Custom GPT

Exploring the advantages and applications in finance.

Multi-Agent Architecture

Designing collaborative AI systems for trading.

Practical Implementation

Translating theory into deployable trading strategies.

Production Challenges

Addressing real-world hurdles and risks.

When to Use Agentic Systems

Identifying optimal scenarios for AI agent deployment.

Lecture Flow

Slide Deck

Overview of Quantitative Trading using AI Agents

Estimated Time:40 **minutes**

Make.com Demo(slide 14)

Practical Application and Platform Walkthrough

Estimated Time: **30 minutes**

Q&A Session

Addressing Your Questions and Insights

Estimated Time: **15 minutes**

Evolution of Quantitative Trading

1. **Manual**
   Human traders executing strategies
2. **Rule-Based Algos**
   Pre-programmed trading rules
3. **ML Models**
   Machine learning predictions
4. **LLMs**
   Large language models for analysis
5. **AI Agents**
   Multi-agent systems working together

Trading research is multi-step, data-heavy, iterative, tool-dependent, and risk-sensitive. Trading is a workflow not a prompt.

The 'Single Prompt' Wall: Why Proximity Prompting Fails at Scale

The "Memory Tax"

Constantly copy-pasting context, brand guidelines, and data into every new chat.

Inconsistency

Getting different results for the same task because the prompt wasn't framed perfectly this time.

The Knowledge Gap

Standard LLMs "hallucinate" when they don't have access to your specific, private documents.

Complexity Fatigue

Prompts are becoming "walls of text" that are hard to manage and easy for the AI to ignore.

Solution: Custom GPT

The Custom GPT: Moving from a 'Chat' to a 'Tool'

A Custom GPT isn't just a saved prompt; it is a pre-configured environment tailored for a specific mission.

Persistent Context

Your instructions are "baked in." No more repeating yourself.

The Knowledge Vault (RAG)

Upload PDFs, spreadsheets, or manuals that the GPT treats as its "Source of Truth."

Systemic Behavior

Define exactly how it should respond (e.g., "Always output in Markdown," "Always ask 3 follow-up questions").

Capability Integration

Toggle on Web Browsing, DALL-E, or Code Interpreter once, and it stays on.

Why Standalone Custom GPTs Fall Short?

While Custom GPTs offer significant advantages over single prompts, their inherent limitations prevent them from scaling to robust enterprise AI applications.

Reliability & Drift

GPTs can **lose context** and custom instructions in long interactions, leading to inconsistent and generic behavior.

Context Collapse

Large knowledge bases cause **"Lost in the Middle"** syndrome, where crucial data is ignored, leading to inaccurate outputs.

Security Risks

Prompt leaking and potential document exposure pose significant threats to sensitive intellectual property and data.

Maintenance Burden

Manual updates for static knowledge bases and **model volatility** (API changes) create constant upkeep challenges.

Isolation & Scalability

Custom GPTs struggle to **collaborate** or integrate seamlessly with external tools, hindering complex workflows.

These critical gaps underscore the need for a more dynamic, coordinated, and resilient approach

Solution: Agentic AI

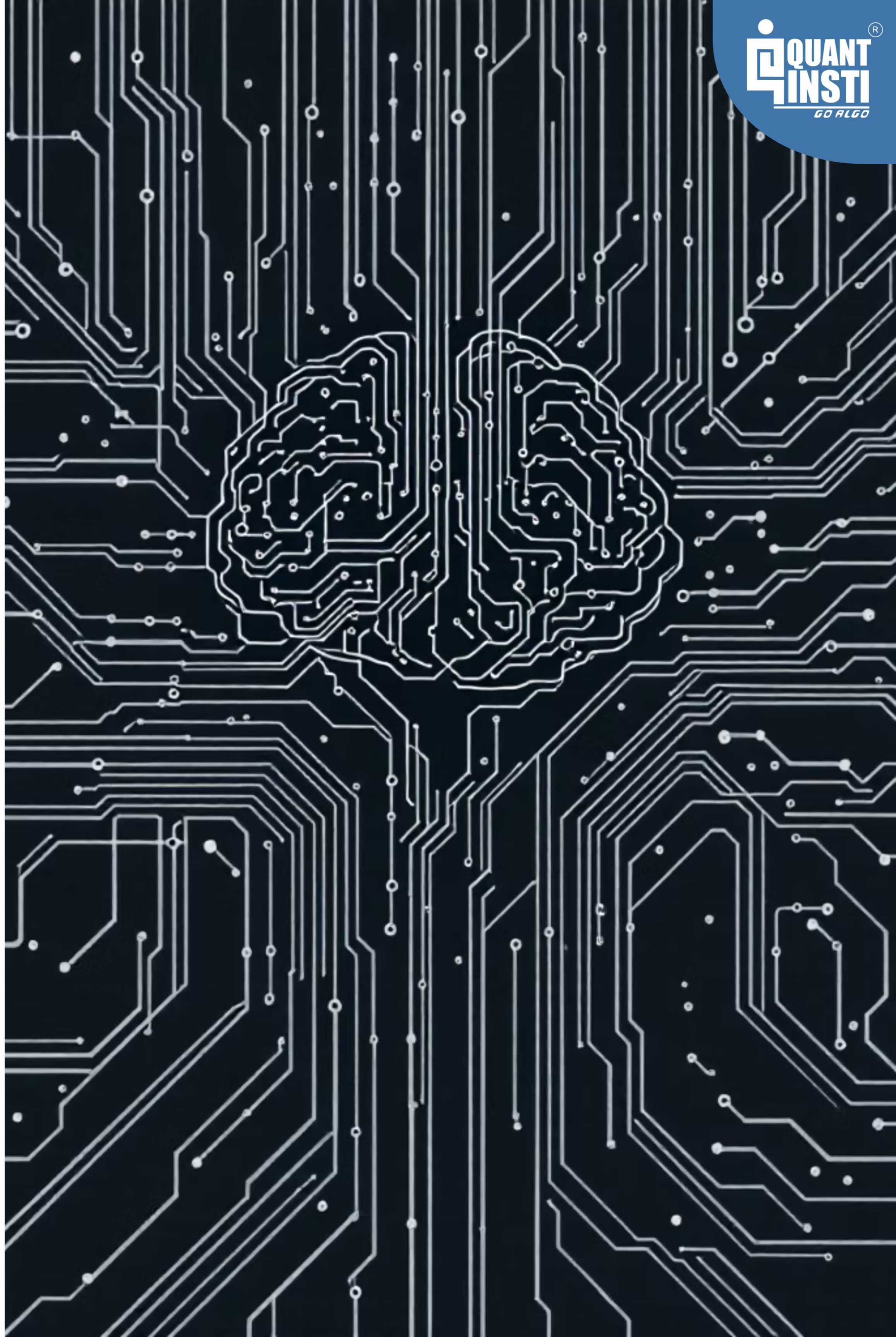
What is Agentic AI?

Key Characteristics

- Goal-driven with clear objectives
- Multi-step planners that break complex problems
- Tool-using with external capabilities
- Stateful with memory across interactions
- Iterative and self-correcting

How Agents Work

1. Plan the approach
2. Act on the plan
3. Evaluate results
4. Revise if needed



Multi-Agent Trading Desk

Research Analyst

Extracts ideas from news and macro data

Data Engineer

Pulls and engineers historical datasets

Quant Developer

Converts ideas into rule logic

Risk Manager

Analyzes exposure and correlations

Portfolio Manager

Monitors performance and drift

Multi-Agent Quant Pipeline

Hypothesis Agent

Extracts ideas from news/macro, defines structured trade thesis, specifies assets & horizon

Data Agent

Pulls historical prices, engineers features, cleans & aligns datasets

Strategy Agent

Converts idea into rule logic, defines entry/exit criteria, sets risk constraints

Backtest Agent

Simulates performance, computes Sharpe/drawdown/turnover, triggers revision if weak

Output: Research-ready dataset → Strategy rules → Performance metrics → Risk analysis → Production deployment

Live Example
Multi-Agent Quant Strategy Workflow

The AI Evolution: From Models to Agents

Three levels of capability, from a general-purpose model to specialist assistants and autonomous agentic workflows.

The Foundation : The Brain

Foundational Model (LLM)

Like a brilliant scholar with broad knowledge.

The Specialist : Subject Matter Expert

Foundation Model + RAG + Custom Instructions

Answers using your data.

The Workforce : Agentic AI

Network of Custom GPTs + Autonomy

Coordinates tools and executes multi-step work.

Production Challenges

Coordination

Hallucinations, conflicting outputs, state misalignment between agents

Cost

Long context windows and iterative loops increase expenses

Reliability

API failures and non-deterministic outputs affect consistency

Governance

Audit logs, explainability, and human-in-the-loop requirements

When to Use Agentic Systems

- Is the workflow multi-step?
- Does it require tool use?
- Is iteration necessary?
- Is state tracking required?
- Is auditability important?

If yes → Agentic architecture adds value. **If no** → Simpler solution may be better.

Questions & Answers

Thank You